MILITARY SPECIFICATION SHEET

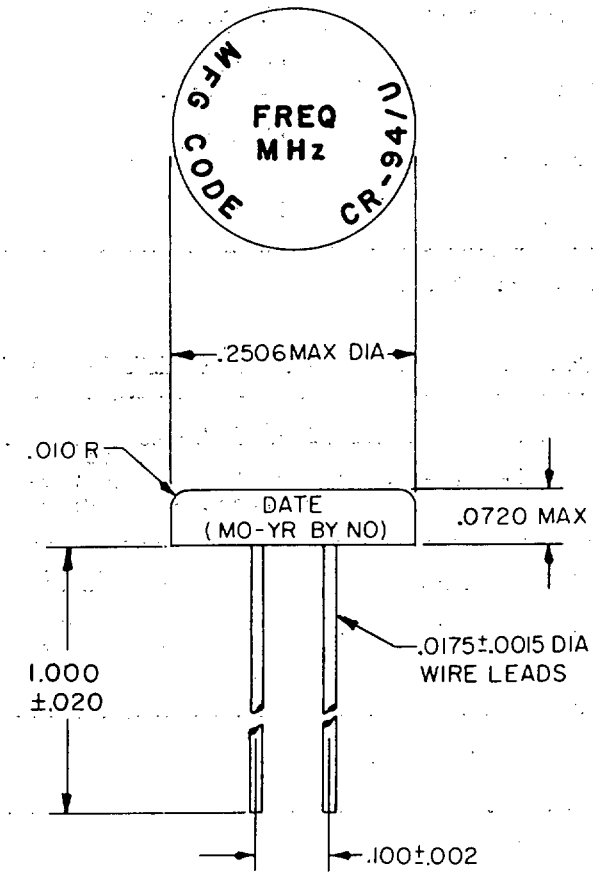
CRYSTAL UNIT, QUARTZ, CR-94/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Ⓑ INACTIVE FOR NEW DESIGN

Pertinent characteristics: 7 to 20 MHz; fundamental; noncontrolled; antiresonance.



INCHES	MM	INCHES	MM
.0015	.04	.072	1.83
.002	.05	.100	2.54
.010	.25	.2506	6.37
.0175	.44	1.00	25.40
.020	.51		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. The type designation, frequency, and code designating characters shall be not less than .035 (.89 mm) in height and not less than .028 (.71 mm) in width.

FIGURE 1. Crystal unit - CR-94/U.

Ⓑ denotes changes

FSC 5955

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:43Z
Check the source to verify that this is the current version before use.

TABLE I. Requirements.

Requirement	Specified value or condition
Holder	Microminiature (see figure 1)
Dimensions, marking, and configuration	See figure 1
Frequency range	7 to 20 MHz, inclusive
Frequency tolerance	±0.005 percent
Antiresonance, load capacitance	30.0 ± 0.5 pF
Equivalent resistance	7 to 15 MHz-according to formula: R = 212.5-12.5 where f is in MHz 15+ to 20 MHz - 25 ohms
Mode of oscillation	Fundamental
Operating temperature range	-55°C to 105°C, inclusive
Test set, calibration values and rated drive level	See table II
Capacitance, shunt	7 pF, maximum
Shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Thermal shock	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	5 ppm

1/ Method 204 of MIL-STD-202, test condition A.

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	MHz	Ohms	mA	Volt	mw
TS-330/TSM	7 to 9	50	--	.32	2.0 ±0.5
	9+ to 12	35	--	.26	2.0 ±0.5
	12+ to 15	25	--	.22	2.0 ±0.5
TS-683/TSM	15+ to 18	15	--	.17	2.0 ±0.5
	18+ to 20	10	--	.14	2.0 ±0.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:43Z
Check the source to verify that this is the current version before use.

Ⓑ QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

Custodian:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5955-0617-2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:43Z
Check the source to verify that this is the current version before use.